Weekly Reflections

Week 15 Updates

During week 15, the following features were implemented:
- Added the Google authentication option to the login page backend logic and frontend UI.
- Altered the seed script to allow for role changes.
- Implemented find-or-link-or-create rule for Google accounts.
- Moved session storage to connect-mongo so sessions survive server restarts.
- Checked my code and env to ensure that the Google OAuth credentials are not committed to GitHub and are properly loaded from environment variables.
- Set HTTP security headers using Helmet.js
- Added logic to validate and sanitize form inputs using express-validator on every POST route that accepts user input
- Added the cookie attribute logic
- Tested the code using automated tests and manual testing in the browser.
- Deployed the final code, drafted the final PRD, and deployed to render.com.

Challenges this week

-Understanding how to run automated tests.
-Understanding how to apply HW15b logic to my site in a way that keeps the admin dash route protected.
- Finishing up and polishing everything that needed to be completed.
- Understanding how to edit the seed script so that the role type could be edited (allowing users to create an account and later request elevated privileges).
- Initially, I wiped my messages and review data by running my seed script. I then had to implement logic so that this wouldn't happen anymore.
- I struggled a bit with formatting my CSS styling for my Google button.
- Making changes to code that allowed it to function using Render. My existing code needed some changes as some of my nav links broke because they were hardcoded to localhost.

Reflection

This week was quite challenging, but very rewarding. I initially struggled to understand how I would implement the logic from HW15b into my code. I then realized that I needed to alter the logic a bit to ensure that only authenticated admins could log in using Google. I then needed to alter my admin seed file to allow privileges/ account types to be changed so that an admin could create an account, and then have their privileges/ role changed using the seed doc. I also added a small message that displays when a new user account is created, stating creation was successful, but the non-admin user account page is under development. In addition, I also made

it so that when the seed script runs, the reviews and messages are not wiped. The testing was also initially a challenge, but once I got into it (utilizing AI tools), I realized that it was fairly straightforward.

All in all, this assignment has been very rewarding. I feel much more knowledgeable about basic syntax and principles. I also better understand how to use AI as a tool to free up time for creative thinking. For example, I was able to implement some complex things that I would not have been able to achieve without the help of an AI agent. This allowed me to be creative and invest time into decision-making, PRD development, and studying the principles taught in this course.

With more time, I would add stronger validation and user feedback for all forms, especially the contact and review forms. I did code the review forum to not accept links, but in a true live site, allowing anyone to publish a review without a validation or review process could be dangerous. I would also improve the admin dashboard design so that messages, reviews, and property updates are easier to manage. I'd also like to add in booking information and maybe a separate page where travelers could look into specifics about the rooms with visuals of all amenities and offerings, with a book now CTA that takes them to an external booking platform.

Week 14 Updates

During week 14, the following features were implemented:
- Created the admin dashboard page with EJS and added a protected route for it in Express.
- Protected admin dashboard route using isAuthenticated
- Added a one-hour session timeout in the existing session config
- Styled the admin login and dashboard pages with CSS
- Added admin header/navigation for the admin pages
- Created the admin special offer editing form on the dashboard
- Ensures that the marketing page's special offer content updates from admin changes
- Added a public review posting form on the marketing page
- Review submission saves into MongoDB and updates the average rating and review count for the property
- Updated the review calculation
- Added a public contact popup form replacing the old mailto logic. Submissions will post on the admin dashboard and save to MongoDB. Admins can clear the messages through a checkbox-based delete function.
- Added an admin "View Reviews" pop-up on the dashboard
- Did some minor UI fixes on the marketing page, like spacing and placeholder text.
- Ensured there was placeholder text on the review form and contact form, and that the review form only accepts ratings between 1 and 5.
- Tested out the password hashing and salting with bcrypt in the admin registration route.
- Original logic allowed people to go back after logging out and view the dashboard. I added logic to prevent this.
- Tested formatting on mobile.

Challenges this week

- Understanding how to implement authentication and protected routes.
- Understanding how to use bcrypt for password hashing and salting.
- Handling some errors I encountered with the new features, such as ensuring the review form only accepts ratings between 1 and 5, and that the contact form submissions are properly saved and displayed on the admin dashboard.
- Implementing the admin features and ensuring that changes made in the admin dashboard properly update the marketing page content.
- CSS styling and layout for the new admin pages and popups. I had an idea of what I wanted, but I needed AI assistance to understand how to implement the design I had in mind.
- General learning curves with new concepts and technologies, such as authentication, bcrypt, and protected routes. The hw14b really helped me understand how to implement these features, though.
- I initially had a hard time understanding how the seed-admin.js logic worked.

Reflection

One challenge I had in integrating the passport.js logic was understanding how it works within the existing application. I was initially confused about how to structure the files and format the code and routes. Because we are using React and Express, I had difficulty conceptualizing how to set up the authentication logic across both the frontend and the backend. I also initially had trouble understanding how the seed-admin.js logic worked. The H14B assignment really set me up for success. I was able to use my HW14B file as a reference and template for how to structure my authentication logic. I also utilized AI tools to understand what changes I needed to make to my existing code to implement the new features. AI tools were very helpful this week, but I also feel like I came away with a better understanding of the concepts. I feel very positive about this week. I had a lot of fun developing the admin dashboard. I actually had to hold back on adding features because I had so many ideas for the admin dashboard and wanted to make sure I had time to implement the core features first. All in all, this week was a great learning experience.

Week 13 Updates

During week 13, the following features were implemented:
- Created the dashboard page using multiple elements: Dashboard.jsx, chart jsx files, component jsx files, data JSON files, dashboard CSS file, marketing page and dashboard page jsx files, arrivals js file, metrics js file, and origin.js file.
- Updated the file structure and code to support a multi-page application.
- Decided on the structure of the site and needed elements using PRD and wireframes.
- Selected which islands I wanted to pull data on.

- Retrieved, formatted, and inserted JSON data (Hawai'i Island, Maui, and Oahu) from DBEDT.
- Added an island selector (slctIsland)
- Added the Arrival Chart, Origin Chart, and Metric Cards elements using API calls and Chart.js.
- Decided on styling elements to be added to dashboard page: live stream, lava video, CTA, misc. strings
- Connected dashboard navigation from the site header.
- Updated the CSS files through multiple prompts as needed using GenAI.
- Researched volcanic data APIs and added VolcWidget. Saw the opportunity to have the background color change according to the aviation code, so I implemented that.
- Updated env: OpenWeatherMap API key in .env as VITE_WEATHER_KEY and tested API calls.
- Ran multiple tests on the app to ensure acceptance criteria are met.
- Ensured that widget updates automatically when the island selector changes.
- Added comments to the code as needed.
- Updated the readme with a guide for users.

Challenges this week

- Understanding how to set up the React project and structure the files.
- Understanding how to transition to a multi-page app.
- Resolving navigation button errors I encountered when paths changed.
- Understanding how to implement the weather widget.
- Understanding "react-router-dom"
- Fetching and formatting the correct data from DBDET.
- Deciding on CSS styling and formatting. Prompting GenAI was somewhat difficult as it struggled to render all of the elements in a pleasing way.
- CSS styling and layout. I wanted to do as much as I could manually, but if I wanted a complex and aesthetically pleasing design, I needed to use some AI assistance to understand how to implement the design I wanted.
- Understanding what makes the widgets update automatically.

Reflection

This week was another challenging week for me. Implementing a large number of changes took quite some time, but I am satisfied with the results, and I better understand what it takes to create a multi-page application. I especially appreciate the knowledge I gained on Chart.js visualizations. In the areas of code that were AI-generated/ assisted, I manually wrote comments in, and this very much helped me better understand the concepts involved. I struggled with the CSS styling a bit this week, as the AI-generated code was not reflecting what I had in mind. Eventually, I remapped my wireframe and was able to get to something satisfactory through prompting and manual adjustments. I also reformatted my marketing page to better align with my brand. All in all, I am satisfied with my dashboard.

Next week, I will implement admin authentication using Passport.js or JWT, create an admin login page, create a protected admin dashboard route, learn how to utlize bcrypt, use csurf middleware to implement CSRF protection on the login, decide on and develop my admin-only features, decide where my special offers and or announcements will post (admin-only feature), and design the layout of my admin page.

Week 12 Updates

During week 12, the following features were implemented:
- I went with option B to prepare for week 13: ensured the local Express server is running and implemented a fetch() call to /api/properties/:id using AI assistance to understand the needed syntax and structure.
- Added all the needed files and updated my database with the new fields for the marketing page.
- Added React functional components and JSX throughout.
- Each major section added as its own component file under src/components/: About, Amenities, Gallery, CTA, Hero, Header, Special Offer, Carousel, and Footer.
- Implemented responsive layout: readable on both desktop and mobile
- Tested routes in Postman. Verified property ID in MongoDB Compass matches the ID in the URL.
- Added alt attributes to all images and ensured color contrast meets WCAG 2.1 AA standards.
- Prompted AI to assist with API calls. Manually coded what I could using Keeper code, W3Schools, Udemy, and other linked React resources.
- Tested local express server and fetch() call in the browser. Verified data is being pulled from the backend and rendered on the marketing page throughout the coding process.
- Added const cors = require("cors"); to server.js and app.use(cors()); to allow cross-origin requests from the React frontend to the Express backend, as fetch was being blocked by CORS policy.
- Added more extensive marketing and property info to my schema and database to align with my PRD. Decided on tagline, amenities, and a package offering.
- Updated README to remove information that goes in the PRD.
- Manually added almost all comments to code (some using autofill from VS code) to increase my understanding and highlight what I manually coded (without AI assistance). Code I manually entered but used AI to help with structuring, syntax, or debugging is not labeled as manually coded, but I did make sure to understand all of the code I entered and the changes I made.
- Designed the wireframe and logo.
- Created property photos using some stock images and some GenAI.
- Manually coded some of my JSX files after I used AI to help me understand how to structure the components (using my local Express server) and what code to use. After the first few sections, I was able to understand the required structure and code and manually implement some sections without AI assistance.

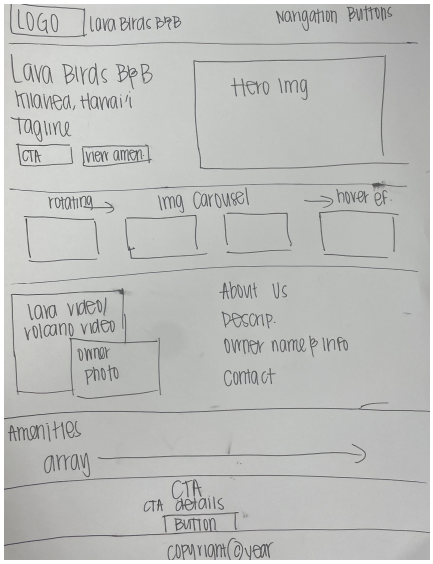
Challenges this week

- Understanding how to set up the React project and structure the files.
- Understanding how to implement the needed changes without hardcoding, as I wanted to go with option B to prepare for week 13.
- Understanding what exactly needed to be done and what changes were feasible for me to make without AI assistance.
- Implementing the fetch() call to pull data from the Express backend and render it on the marketing page.
- Ensuring the marketing page is responsive and meets accessibility standards.
- Navigating the issues I encountered with the CORS policy and understanding how to resolve it by allowing cross-origin requests in the Express server.
- Troubleshooting MongoDB Compass to verify the property ID and ensure it matches the ID in the URL for the fetch() call.
- Figuring out how to add navigation links to my header.jsx file.
- CSS styling and layout. I wanted to do as much as I could manually, but if I wanted a complex and aesthetically pleasing design, I needed to use some AI assistance to understand how to implement the design I wanted.

Reflection

This week was very challenging but also very rewarding. I implemented all of the features above in my term project application. Although I did use AI tools, I began by creating all my files using Keeper code and Udemy as a reference to manually add what I could. As I went with option B, I needed some assistance with syntax. I decided to manually add/copy over all AI-generated code instead of prompting with Codex in VS Code for automated changes, and it really helped to increase my understanding of React. After a while, I began to better understand how to structure JSX files and was able to do some minor coding myself without any assistance. I also changed the information in my database to apply to the new one-property structure. I struggled a bit when it came to debugging and error handling, but I was able to use DevTools and AI assistance to figure out any errors I encountered. I did some back-and-forth prompting when it came to CSS styling, as I had a vision in mind that I could not feasibly accomplish without AI tools. I would write lengthy prompts, go over the code, copy the code over, run the program, and then start again. I am still not 100% satisfied with my styling and will probably change the styling a bit more next week as I add in my dashboard. Wireframing really helped me, so I will likely do the same next week. I also had fun making a Logo in Canva. I tried to find existing images online, but ended up using GenAI to create property images, which was more time consuming that I expected. It ended up being satisfying, though, as I could really tailor the images to apply to what was in my PRD. The PRD was also especially helpful this week as a planning tool.

Wireframe



Logo First Draft

Final Logos

Week 11 Updates

During week 11, the following features were implemented:
- Review schema (guestName, rating 1–5, comment, date) embedded in the Property document.
- Express routes: GET /properties, GET /properties/:id, POST /properties/:id/reviews.
- Mongoose query operators ($gte, $lte) added to filter properties by minimum/maximum rating.
- All routes tested in Postman.
- Postman collection exported and committed to the repo.

- Basic EJS template rendered at /properties listing all properties.
- Checked EJS rendering in browser.

Looking to week 12, the following features are ready to be implemented:
- React Marketing Page
  - Route is prepared in properties.js.
  - Endpoint returns all fields the marketing page will need: name, island, type, description, amenities, targetSegment, imageURL.
  - It will be important to meet the required sections, use React functional components, apply responsive CSS, and validate WCAG 2.1 AA compliance.
  - The features below are proposed for the marketing page.

Challenges this week

- Understanding how to structure the review schema and embed it in the Property document.
- Navigating Postman to test the new routes and ensure they were working as expected. Now understood.
- Understanding ($gte, $lte) operators and how to implement them for filtering by ratings